



Federal Board HSSC-I Examination

Physics Model Question Paper

Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)

BY FAKHAR STEM SPHERE

Section - A (Marks 17)

Time Allowed: 25 minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

Model Paper

ROLL NUMBER					

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Version No.			

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Candidate Sign. _____

Invigilator Sign. _____

Q1. Fill the relevant bubble against each question according to curriculum. Each part carries one mark.

S #	Question	(A)	(B)	(C)	(D)	(A)	(B)	(C)	(D)
i.	Which of the following quantities are dimensionally similar?	Torque and work	Stress and spring constant	Angle and length	Strain and pressure	☐	☐	☐	☐
ii.	Maximum work is obtained by the process called:	Isochoric	Isothermal	Isobaric	Adiabatic	☐	☐	☐	☐
iii.	Projectile motion is two-dimensional motion under constant acceleration due to	Air resistance	Gravity	Horizontal distance	Velocity	☐	☐	☐	☐
iv.	A turtle travel 45cm in 20 minutes, its average velocity will be:	2.25cm/h	90cm/h	135cm/h	15cm/h	☐	☐	☐	☐
v.	A stationary wave is formed by waves of frequency 512Hz. The speed of the wave is 65m/s, the distance between two consecutive antinodes is:	0.126m	0.06m	0.03m	0.024	☐	☐	☐	☐
vi.	A wire is stretched to double of its original length, its strain is	0.5	1	zero	2	☐	☐	☐	☐
vii.	Cyclotron frequency of a charged particle moving in a magnetic field B is independent of:	Magnetic flux density	Charge on a particle	Velocity of particle	Mass of particle	☐	☐	☐	☐
viii.	The best instrument for accurate measurement of emf of a cell is:	Voltmeter	Potentiometer	Ammeter	Ohmmeter	☐	☐	☐	☐
ix.	What is the base unit of pressure?	kgms ⁻¹	kgm ⁻¹ s ⁻¹	kgm ⁻¹ s ⁻²	kgm ² s ⁻²	☐	☐	☐	☐
x.	Theory of relativity was formulated by.	Isaac Newton	Rutherford	Plank	Albert Einstein	☐	☐	☐	☐
xi.	The radius at two ends of a pipe is in the ratio of 2:3, then the speed of liquid at the two ends is in the ratio of:	2:3	3:2	4:9	9:4	☐	☐	☐	☐
xii.	Lepton number for muon particle is:	-1	+1	+1/2	0	☐	☐	☐	☐

xiii.	Which of the following moving particle is not deflected by magnetic field?	Neutron	Proton	Electron	Positron	☐ ☐ ☐ ☐
xiv.	The force that acts on the body but it does no work is called:	Frictional force	Gravitational force	Centripetal force	Elastic force	☐ ☐ ☐ ☐
xv.	The region around a charge in which a test charge can feel an electric force is called	Electric intensity	Electric Potential	Electric Field	Electric Energy	☐ ☐ ☐ ☐
xvi.	Moment of inertia of a spinning body about its axis does not depend on the:	Angular velocity of body	Mass of body	Orientation of the axis	Distribution of the mass around axis	☐ ☐ ☐ ☐
xvii.	When $F_x = 3 \text{ N}$ and $F = 5 \text{ N}$ then $F_y =$	6N	8N	2N	4N	☐ ☐ ☐ ☐



PHYSICS HSSC-I

SECTION – A (Marks 17)

Time allowed: 25 Minutes

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حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر باقی مرکز کے حوالے کریں۔ گات کر دہا،
لکھنے کی اجازت نہیں ہے۔ سیاہ قلم استعمال نہیں ہے۔

1st Annual 25 Local

Version No.				
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ROLL NUMBER						

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①	①	①	①	●
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⑨	⑨	⑨	⑨	⑨	⑨	⑨

Answer Sheet No. _____

بر سوال کے سامنے دیے گئے ہر کولم کے مطابق درست دائرہ کو پر کریں۔

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. The dimensions $[ML^2T^{-2}]$ show:	Power	Work	Velocity	Acceleration	☐	☐	☐	☐
2. If $r = 2.25 \pm 0.01 \text{ cm}$ then percentage uncertainty in " r " will be:	0.2%	0.3%	0.4%	0.5%	☐	☐	☐	☐
3. $\hat{j} \cdot (\hat{i} \times \hat{k}) =$	0	-1	1	2	☐	☐	☐	☐
4. Magnitude of centripetal force $F_c =$	$mr\omega^2$	$mr\omega$	$n\omega^2$	mv^2	☐	☐	☐	☐
5. If the length of blade of a fan is increased, then its rotational speed (at same voltage and current):	Will increase	Will decrease	Will become infinite	Remains constant	☐	☐	☐	☐
6. Two bodies A and B of same masses have velocities in the ratio 2:1, their kinetic energies are in the ratio:	2:1	1:2	4:1	1:4	☐	☐	☐	☐
7. Maximum drag force on a 1kg falling sphere is:	9.8 N	1 N	98 N	4.9 N	☐	☐	☐	☐
8. Young's Modulus, $Y =$	$\frac{\Delta L / L}{F / A}$	$\frac{FA}{\Delta L L}$	$\frac{\Delta L L}{FA}$	$\frac{F / A}{\Delta L / L}$	☐	☐	☐	☐
9. In an adiabatic process which quantity remains constant?	Internal energy	Volume	Temperature	Mass	☐	☐	☐	☐
10. Location of a submarine in deep sea can be detected by:	Doppler effect	Compton's effect	Faraday's Law	Bernoulli principle	☐	☐	☐	☐
11. An electron is placed in a uniform electric field directed from south to north. In which direction will electron move after it is released?	East	South	North	West	☐	☐	☐	☐
12. In order to achieve high accuracy, the slide wire of a potentiometer should be:	As long as possible	As short as possible	Very thick	Neither too small, nor too large	☐	☐	☐	☐
13. Two closely placed parallel wires are carrying current in opposite direction, they will:	Repel each other	Attract each other	Not affect each other	Rotate each other	☐	☐	☐	☐
14. One weber is equal to:	NA^{-1}	NmA^{-1}	$Nm^{-1}A$	$Nm^{-1}A^{-1}$	☐	☐	☐	☐
15. When an object moves with a very high speed its length _____ in the direction of its motion.	Contracts	Expands	Remains same	Becomes zero	☐	☐	☐	☐
16. Relativistic mass is always:	Less than rest mass	Greater than rest mass	Equal to rest mass	Zero	☐	☐	☐	☐
17. A hadron may either be a Baryon or a:	Lepton	Gluon	Photon	Meson	☐	☐	☐	☐

— 1HA-I 25004 (B) —

$$\vec{A} \cdot \vec{B} = AB \cos \theta \quad L = I\omega \quad K.E = \frac{1}{2}mv^2 \quad W = mg \quad g = 9.8ms^{-2} \quad v = r\omega \quad \vec{A} \times \vec{B} = AB \sin \theta \quad \phi_B = \vec{B} \cdot \vec{A} \quad \text{uncertainty} = \frac{\text{Absolute uncertainty}}{\text{No. of vibrations}}$$



PHYSICS HSSC-I SECTION – A (Marks 17)

Time allowed: 25 Minutes

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حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر نام مرکز کے حوالے کریں۔ کٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ سب سے پہلے کا استعمال ممنوع ہے۔

*1st Annual 25
Hard*

Version No.				
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ROLL NUMBER						

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Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. In a simple pendulum experiment, percentage errors (uncertainty) in "L" and "T" are 0.1% and 2% respectively. Maximum percentage error (uncertainty) in "LT ² " is:	4.1%	3%	2.1%	3.2	☐	☐	☐	☐
2. If $\vec{A} \cdot \vec{B} = \vec{A} \times \vec{B} $ then angle "θ" between vectors $\vec{A}$ and $\vec{B}$ is:	0	$\pi/4$	$\pi/2$	π	☐	☐	☐	☐
3. When a massive body of mass M ₁ is moving with velocity V ₁ collides with lighter stationary body of mass m ₂ , the velocity of massive body after collision will be:	$V_1' = 2V_1$	$V_2' = V_1$	$V_1' = V_1$	$V_2' = 2V_2$	☐	☐	☐	☐
4. Moment of inertia of a body spinning about its axis DOES NOT depend on the:	Angular velocity of body	Mass of body	Orientation of the axis	Distribution of mass around axis	☐	☐	☐	☐
5. The bodies A and B of equal masses have velocities ratio in 2:1 their K.E are in the ratio of:	2:1	1:2	4:1	1:4	☐	☐	☐	☐
6. The fluid is said to be incompressible, if its density is:	Constant	Very high	Very small	Zero	☐	☐	☐	☐
7. Equation of continuity is based upon law of conservation of:	Momentum	Energy	Mass	Charge	☐	☐	☐	☐
8. Glass is an example of _____ solid.	Crystalline	Amorphous	Poly-crystalline	Ductile	☐	☐	☐	☐
9. Young's Modulus, Y=	$\frac{\Delta L / L}{F / A}$	$\frac{FA}{\Delta L L}$	$\frac{\Delta L L}{FA}$	$\frac{F / A}{\Delta L / L}$	☐	☐	☐	☐
10. A Carnot engine undergoes _____ processes in one cycle.	Two	Three	Four	Six	☐	☐	☐	☐
11. Stars moving away from the earth show:	Red shift	Blue shift	Black shift	Yellow shift	☐	☐	☐	☐
12. A negatively charged particle is placed in a uniform electric field directed from south to north. In which direction will the particle move after being released?	North	South	East	West	☐	☐	☐	☐
13. A 100 ohm resistor is connected across the terminals of a 10V battery. The power dissipation of the resistor is:	1 watt	10 watts	100 watts	1000 watts	☐	☐	☐	☐
14. The SI unit of magnetic induction is:	Nm	Gauss	Weber	Tesla	☐	☐	☐	☐
15. Relativistic mass is always:	Less than rest mass	Greater than rest mass	Equal to rest mass	Zero	☐	☐	☐	☐
16. Inertial frame of reference is a frame in which _____ law of motion holds.	1 st	2 nd	3 rd	4 th	☐	☐	☐	☐
17. Two up quarks and one down quark make a:	Meson	Photon	Neutron	Proton	☐	☐	☐	☐

1HA-I 25004 (D)

• $(K.E)_{max} = \frac{1}{2}mv^2$ • $l = mr^2$ • $P = VI$ • $P = \frac{V^2}{R}$ • $\vec{A} \cdot \vec{B} = AB \cos \theta$ • $\vec{A} \times \vec{B} = AB \sin \theta$ • $L = I\omega$ • $v = r\omega$



PHYSICS HSSC-I

SECTION – A (Marks 17)

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2nd Annual 25

Version No.				
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ROLL NUMBER						

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6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. The unit of force in terms of base units:	kg-m/s	kg-m/s ²	kg-m ² /s	kg ² -m/s ²	☐	☐	☐	☐
2. The dimension $[ML^{-1}T^{-2}]$ represents:	Pressure	Force	Power	Work	☐	☐	☐	☐
3. Torque $\vec{\tau}$ on a body is given by:	$\vec{\tau} = \vec{m} \times \vec{a}$	$\vec{\tau} = \vec{F} \times \vec{d}$	$\vec{\tau} = \vec{r} \times \vec{F}$	$\vec{\tau} = \vec{l} \times \vec{\omega}$	☐	☐	☐	☐
4. For a freely falling body, the vertical velocity at the 4 th second is:	94.9 ms ⁻¹	47.5 ms ⁻¹	40 ms ⁻¹	39.2 ms ⁻¹	☐	☐	☐	☐
5. The maximum range of a projectile is obtained when launch angle $\theta =$	30°	45°	60°	90°	☐	☐	☐	☐
6. The SI unit of angular velocity is:	Rad	N-m	m/s ²	rad/s	☐	☐	☐	☐
7. The centripetal force for a body of mass m moving with speed v in a circle of radius r is:	mv	mv^2r	$\frac{mv^2}{r}$	$\frac{m}{r^2}$	☐	☐	☐	☐
8. The work done by a gas at constant pressure is given by:	$P \cdot \Delta V$	$\frac{\Delta V}{P}$	ΔV	$V \cdot \Delta P$	☐	☐	☐	☐
9. The first law of thermodynamics is expressed as:	$Q = U + V$	$Q = T + S$	$Q = \Delta U + W$	$Q = mL$	☐	☐	☐	☐
10. The Doppler effect explains the change in:	Frequency	Amplitude	Speed of light	Wavelength only	☐	☐	☐	☐
11. For constructive interference, path difference Δx should be:	$n\lambda$	$\frac{(2n+1)\lambda}{2}$	$\frac{(n+1)\lambda}{2}$	$\frac{n\lambda}{2}$	☐	☐	☐	☐
12. Electric field intensity E is defined as:	F/d	F/q	V/l	W/q	☐	☐	☐	☐
13. A 10 Ω resistor is connected across the terminals of a 10V battery. The power dissipation of the resistor is:	100 W	50 W	10 W	25 W	☐	☐	☐	☐
14. Magnetic flux Φ through a surface area A is:	B/A	$B \cdot A$	A/B	B/A^2	☐	☐	☐	☐
15. Lenz's law states that induced emf opposes:	Current	Resistance	Power factor	Change causing it	☐	☐	☐	☐
16. An inertial frame of reference is the one:	Which is at rest	Which has zero acceleration	Which is under a force	Which has variable acceleration	☐	☐	☐	☐
17. At what speed the mass of a body becomes double of its proper mass?	$\frac{2}{3}c$	$\frac{\sqrt{3}}{2}c$	$\frac{\sqrt{2}}{3}c$	$\frac{1}{3}c$	☐	☐	☐	☐

— 1HA-II 25004 —

$$P = \frac{V^2}{R} = I^2 R = V \times I$$

$$g = 9.8 \text{ ms}^{-2}$$

$$F = ma$$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$v_f = v_i + gt$$